Speech, Language & Dysarthria Clinical Assessment Summary

Report ID: SC-ASSESS-000003 | Assessment Date: October 27, 2026 at 04:14 UTC | Patient ID: SC-PT-1001

Patient Identifier:	Sarah Jenkins (SC-PT-1001)	**Age & Biological Gender:**	48 yrs / Female
Neurological History:	Stroke, Hypertension	**Symptom Onset / Duration:**	Sudden (<24h)
Prior Assessments:	Frenchay Dysarthria Assessment-2 (FDA-	**Therapy / Treatment History:**	Outpatient Speech-Language The
Language / Dialect:	English (US)	**Communication Needs:**	Mild vocal fatigue in late aft

1. Audio Quality & Recording Validation

Quality Status:	GOOD	**Signal-to-Noise Ratio:**	26.5 dB	**Clipping Check:**	Clear (No clipping)
Evaluated Task:	Sustained Vowel	**Speech Duration:**	2.5s total (2.5s active speech)		

2. Dysarthria Severity Estimation & Speech Production Analysis

ESTIMATED SEVERITY: NO CLEAR DYSARTHRIC INDICATORS DETECTED **Calibrated Confidence:** 1.0%

Intelligibility Index:	97.4% (Estimated)	**Speech Rate:**	250.0 WPM (6.8 syl/sec)
Articulation Accuracy:	98.4/100 (Crisp & Precise Cons)	**Vocal Quality:**	Normal Vocal Resonance (Clear &
Prosody / Intonation:	Monopitch / Monoloudness (	**Respiration Coordination:**	Adequate Subglottic Breath Suppo

Contributing Features: Acoustic biomarkers (Jitter, Shimmer, HNR) fall within normative physiological ranges

Notice: Severity estimate reflects acoustic screening metrics and is NOT a definitive clinical diagnosis.

3. Screening for Associated Speech & Language Disorders (Differential Profile)

Condition Screened	Screening Index	Clinical Distinction	Evaluation Status
Apraxia of Speech (AOS):	5.0%	Motor Planning vs Execution	Screening Indicator (Not a Diagnosis)
Aphasia (Language):	5.0%	Linguistic Formulation / Retrieval	Requires Bedside Language Battery
Stuttering / Disfluency:	5.0%	Speech Fluency Breaks & Blocks	Disfluency Frequency Analysis Required

Differential distinction: Dysarthria (neuromuscular execution) vs Apraxia (motor planning) vs Aphasia (linguistic).

4. Measured Acoustic Biomarkers & Physical Signal Properties

Acoustic Biomarker	Measured Value	Reference Norms	Physiological Interpretation
Acoustic Speaker Gender	Female (70.6%)	Pitch F0 & Centroid	Vocal tract acoustic profile
Mean Pitch (F0)	175.6 Hz	M: 85-165Hz | F: 165-260Hz	Vocal fold vibrational frequency
Jitter (Local Perturbation)	0.20%	< 1.50% (Normal)	Pitch period vibration stability
Shimmer (Amplitude Perturbation)	4.86%	< 4.00% (Normal)	Glottal resistance & amplitude
Harmonics-to-Noise Ratio (HNR)	16.4 dB	> 15.0 dB (Normal)	Acoustic periodicity vs breathiness
Silence / Pause Dynamics	15.0%	10% - 25% (Typical speech)	Articulatory fluency & hesitations

6. Clinical Next Steps & Professional Evaluation Plan

* Formal Motor Speech Evaluation by a licensed CCC-SLP (e.g., Frenchay Dysarthria Assessment-2 or Apraxia Battery for Adults-2).
* Comprehensive Language Assessment (e.g., Western Aphasia Battery-Revised or Boston Diagnostic Aphasia Examination) if word retrieval or syntax difficulties persist
* Laryngeal Videostroboscopy if dysphonia, vocal tremor, or vocal fold bowing is suspected.

7. Clinician Review & Attestation Section

Review Status: PENDING FORMAL CLINICAL REVIEW **Reviewing Clinician: Not Assigned**

Clinician Notes: No clinician review notes recorded yet.

MANDATORY CLINICAL SCREENING & UNCERTAINTY DISCLAIMER:

SCREENING SUPPORT ONLY. REQUIRES FORMAL EVALUATION BY A CERTIFIED SPEECH-LANGUAGE PATHOLOGIST.

SpeechCare AI is an assistive screening and clinical decision-support prototype. It does NOT provide a definitive medical diagnosis. All screening indicators, severity classifications, and acoustic indices account for potential limitations in audio quality, microphone hardware, and environmental noise. These findings must NEVER replace a comprehensive bedside evaluation by a certified Speech-Language Pathologist (CCC-SLP) or licensed neurologist. Further clinical assessment is strongly advised before therapeutic or pharmacological interventions.